

Homework 3: One-Sample Tests and ANOVA

Submission Notes: Please write your answers to each question under that question (not altogether at the end). It will be easier for me to give full credit if your answer is commented well, including comments about where you are completing each sub-part of the question. For an example, see Homework 1.

1. On Homework 2, you developed an exact test for $E(X) = \mu_X = \lambda$ for data drawn from an exponential distribution with mean λ .

For context, we did this because the Central Limit Theorem,

$$T = \frac{\bar{X} - \lambda}{s/\sqrt{n}} \sim \mathcal{AT}(v = n - 1),$$

can only be used when n is large. That is, this approximation is “good” when n is large. However, the *exact* sampling distribution of $\bar{X}$,

$$\bar{X} \sim \text{Gamma}\left(\alpha = n, \beta = \frac{n}{\lambda}\right),$$

works regardless of n .

When asked to compute a p -value, many of you used the information you found using a large language model. Namely, you used the suggested p -value calculation

$$p\text{-value} = 2 \times \min(F_{\bar{X}}(\bar{x}), 1 - F_{\bar{X}}(\bar{x}))$$

instead of simply directly computing

$$\begin{aligned} p\text{-value} &= P(\bar{X} \leq \lambda_0 - |\lambda_0 - \bar{x}|) + P(\bar{X} \geq \lambda_0 + |\lambda_0 - \bar{x}|) \\ &= P(\bar{X} \leq \lambda_0 - |\lambda_0 - \bar{x}|) + (1 - P(\bar{X} \leq \lambda_0 + |\lambda_0 - \bar{x}|)) \\ &= F_{\bar{X}}(\lambda_0 - |\lambda_0 - \bar{x}|) + (1 - F_{\bar{X}}(\lambda_0 + |\lambda_0 - \bar{x}|)). \end{aligned}$$

My professor once told me “you don’t have to remember anything, you can just derive what you need.” To be equally annoying, I will retool this to say “you don’t have to blindly trust anything, you can simulate it to check!”

- (a) Conduct a simulation to evaluate the probability of Type I error using both approaches by completing the following in a loop. Report the resulting Type I error estimate.
 - i. Generate a random sample of $n = 10$ observations from an exponential distribution with mean $\lambda = 2$.
 - ii. Compute $\Delta = |\bar{x} - \lambda_0|$, where $\lambda_0 = 2$.
 - iii. Compute the lower part of the p -value.

$$p_L = P(\bar{X} \leq \lambda_0 - \Delta)$$

- iv. Compute the upper part of the p -value.

$$p_U = P(\bar{X} \geq \lambda_0 + \Delta)$$

- v. Compute the exact p -value.
- vi. Compute the p -value suggested by the large language model.

```
set.seed(131313)

exact_rejection_rate <- rep(NA, 10000)
LLM_rejection_rate <- rep(NA, 10000)

n = 10
```

```

lambda = 2
lambda0 = 2

for (i in 1:10000){

  # part i
  rsamp <- rexp(n = n, rate = 1/lambda)

  # part ii
  delta = abs(mean(rsamp) - lambda0)

  #part iii
  pL = pgamma(q = lambda0-delta, shape = n, rate = n/lambda0, lower.tail = T)

  # part iv
  pU = pgamma(q = lambda0+delta, shape = n, rate = n/lambda0, lower.tail = F)

  #part v
  exact_p = pL + pU
  exact_rejection_rate[i] = ifelse(exact_p < 0.05, 1, 0)

  #part vi
  LLM_lower = pgamma(mean(rsamp), shape = n, rate = n/lambda0)
  LLM_upper = 1 - pgamma(mean(rsamp), shape = n, rate = n/lambda0)
  LLM_p = 2*min(LLM_lower, LLM_upper)
  LLM_rejection_rate[i] = ifelse(LLM_p < 0.05, 1, 0)

}

(mean(exact_rejection_rate))

## [1] 0.0498

(mean(LLM_rejection_rate))

## [1] 0.0479

```

The α of the exact method is 0.0498. The α of the LLM approach is 0.0479

- (b) Conduct a simulated power analysis to evaluate the power and probability of Type II error using both approaches by completing the following in a loop. Report the resulting Power estimate.
- Generate a random sample of $n = 10$ observations from an exponential distribution with mean $\lambda = 2.3$.
 - Compute $\Delta = |\bar{x} - \lambda_0|$, where $\lambda_0 = 2$.
 - Compute the lower part of the p -value.

$$p_L = P(\bar{X} \leq \lambda_0 - \Delta)$$

- Compute the upper part of the p -value.

$$p_U = P(\bar{X} \geq \lambda_0 + \Delta)$$

- Compute the exact p -value.
- Compute the p -value suggested by the large language model.

```

exact_power <- rep(NA, 10000)
LLM_power <- rep(NA, 10000)

n = 10
lambda = 2.3
lambda0 = 2

for (i in 1:10000){

  # part i
  rsamp <- rexp(n = n, rate = 1/lambda)

  # part ii
  delta = abs(mean(rsamp) - lambda0)

  #part iii
  pL = pgamma(q = lambda0-delta, shape = n, rate = n/lambda0, lower.tail = T)

```

```

# part iv
pU = pgamma(q = lambda0+delta, shape = n, rate = n/lambda0, lower.tail = F)

#part v
exact_p = pL + pU
exact_power[i] = ifelse(exact_p < 0.05, 1, 0)

#part vi
LLM_lower = pgamma(mean(rsamp), shape = n, rate = n/lambda0)
LLM_upper = 1 - pgamma(mean(rsamp), shape = n, rate = n/lambda0)
LLM_p = 2*min(LLM_lower, LLM_upper)
LLM_power[i] = ifelse(LLM_p < 0.05, 1, 0)
}

(mean(exact_power))
## [1] 0.1125
(mean(LLM_power))
## [1] 0.0853

```

The power of our exact method is 0.1125. The power of the LLM approximation is 0.0853.

- (c) To gain insight into your results, compute the exact p -value and the p -value suggested by the large language model for a fine grid of values of Δ from 0 to 4 for $\lambda_0 = 2$. Plot the resulting p -values against a $y = x$ line. Explain how the visual maps onto the results of the simulations.

```

lambda0 = 2
n = 10

delta <- seq(0,4,length.out = 1000)

exact_p <- rep(NA, 1000)

LLM_p_high <- rep(NA, 1000)
LLM_p_low <- rep(NA, 1000)

for(i in 1:1000){

  #exact p-value calculation
  exact_lower <- pgamma(lambda0-delta[i], shape = n, rate = n/lambda0, lower.tail = T)
  exact_upper <- pgamma(lambda0+delta[i], shape = n, rate = n/lambda0, lower.tail = F)
  exact_p[i] = exact_lower + exact_upper

  #LLM p-value calculation
  xbar_approx_low = lambda0-delta[i]
  xbar_approx_high = lambda0 + delta[i]
  LLM_p_low[i] = 2 * pgamma(xbar_approx_low, shape = n, rate = n/lambda0, lower.tail = T)
  LLM_p_high[i] = 2 * pgamma(xbar_approx_high, shape = n, rate = n/lambda0, lower.tail = F)

}

library(tidyverse)
library(ggplot2)

p_values <- tibble(exact = exact_p,
                   LLM_low = LLM_p_low,
                   LLM_high = LLM_p_high)

ggplot(data = p_values) +
  geom_point(aes(x = exact, y = LLM_low, color = "Low Estimate")) +
  geom_point(aes(x = exact, y = LLM_high, color = "High Estimate")) +
  geom_abline(intercept = 0, slope = 1, color = "black", linetype = "dashed") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  xlab('Exact p-value') +
  ylab('LLM p-value') +
  ggtitle('Exact vs. LLM Approach') +
  theme(legend.title=element_blank()) +
  xlim(0,1) +
  ylim(0,1)

```

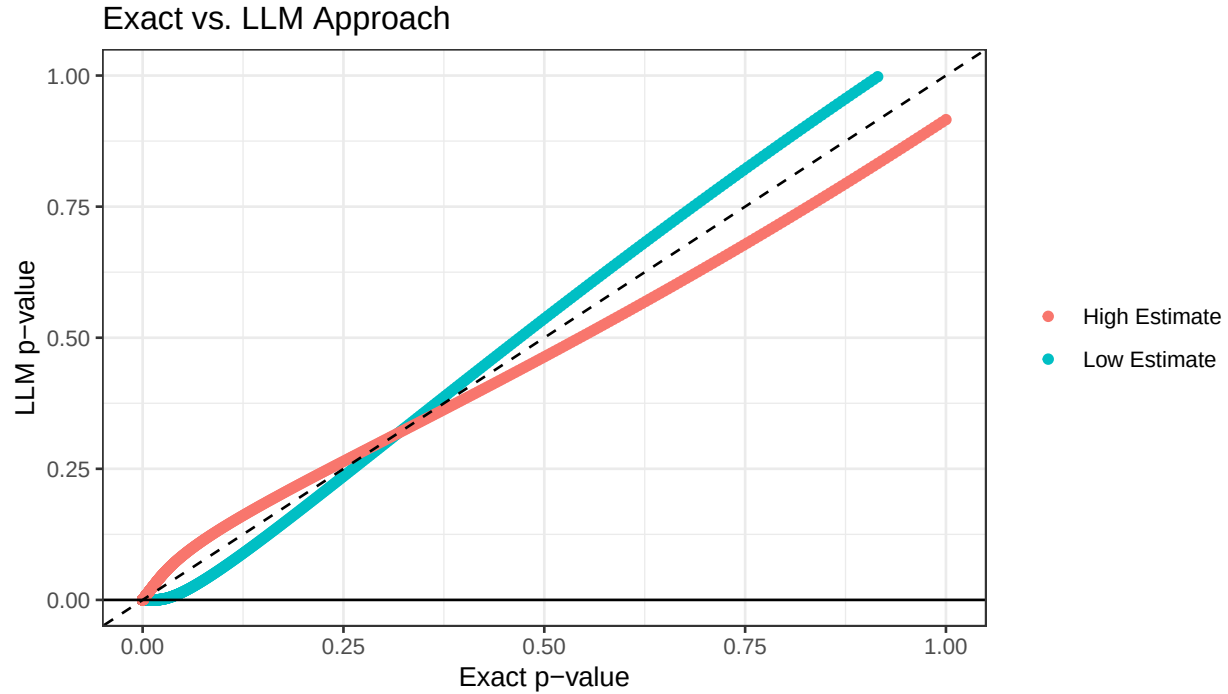


Figure 1: p-value Comparison

- (d) Rerun the simulations when $n = 1000$. Just rerun your code above with the sample size changed and report the results here – no need to copy and paste the code again here. Provide the results and comment on how they have changed from $n = 10$. What did students who used the large language model unwittingly assume, and is it problematic?

As the sample size increases to $n = 1000$, the two methods are virtually identical in terms of α and Power. Students who used the LLM version of the calculation assumed that the LLM method would be approximately robust even with a low sample size, as the exact version is. This can be problematic as most datasets used in statistics are not as large as $n = 1000$, so the difference in α and power could make a difference in the hypothesis test.

2. Under the alternative hypothesis, the distribution of the ANOVA F statistic follows a non-central F distribution. An added non-centrality parameter (λ) shifts the mean of the usual F distribution away from 1 (what we expect under H_0). Use this fact to conduct a power analysis for the ANOVA test.

- (a) Let $N = 542$ and $k = 5$ based on the MFAP4 example from class. Find the F statistic that starts the rejection region for the usual ANOVA test.

Hint: The degrees of freedom are $df_1 = k - 1$, $df_2 = N - k = k(n - 1)$ when the sample sizes are the same.

```
N = 542
k = 5

(rejection_statistic <- qf(0.95, df1 = k-1, df2 = N-k, lower.tail = T))

## [1] 2.388533
```

- (b) Convert the effect size of the ANOVA test – $\omega^2 = 0.23$ for the MFAP4 data – to Cohen's f using the following formula.

$$\text{Cohen's } f = \sqrt{\frac{\omega^2}{1 - \omega^2}}$$

```
omega2 <- 0.23

(cohensf <- sqrt(omega2/(1-omega2)))

## [1] 0.5465357
```

- (c) Let $N = 542$ and $k = 5$ based on the MFAP4 example from class. Use Cohen's f from the previous part to compute the non-centrality parameter.

$$\begin{aligned}\lambda &= N \times (\text{Cohen's } f)^2 \\ &= k \times n \times (\text{Cohen's } f)^2\end{aligned}\quad (\text{Equal sample size } n)$$

```
(lambda = N * cohensf^2)

## [1] 161.8961
```

- (d) Let $n = 542$ and $k = 5$ based on the MFAP4 example from class. What is the probability that we observe the F statistic in (a) under the F distribution with $df_1 = k - 1$, $df_2 = N - k = k(n - 1)$ or one that provides more support for the alternative when the sample sizes are the same, and non-centrality parameter λ from the previous part?

```
n = 542
k = 5

pf(q=rejection_statistic, df1 = k-1, df2 = n-k, ncp = lambda, lower.tail = F)

## [1] 1
```

- (e) Write a function that takes k , and ω^2 and computes the power:

$$P(F \geq F_{k-1, k(n-1), \lambda=0} | k-1, k(n-1), \lambda = \lambda).$$

Hint: A loop should be helpful here.

```
power <- function(k, omega2, desired_power){

  cohensf <- sqrt(omega2/(1-omega2))

  n = 3

  repeat{

    rejection <- rep(NA, 1000)

    lambda = n * k * cohensf^2

    rejection_stat = qf(0.95, df1 = k-1, df2 = k*(n-1))

    p = pf(q = rejection_stat, df1 = k-1, df2 = k*(n-1), ncp = lambda, lower.tail = F)

    if(p >= desired_power){
      return(n)
      break
    }else{
      n = n + 1
    }
  }
}
```

- (f) Use the function in the previous part to find what n has to be to ensure this probability is 80% when $k = 5$ and $\omega^2 = 0.14$ (a moderate effect)?

Hint: You can check your answer using `pwr::pwr.anova.test()`.

```
power(k = 5, omega2 = 0.14, desired_power = 0.8)

## [1] 16

pwr::pwr.anova.test(k = 5, power = 0.8, f = sqrt(0.14/(1-0.14)))
```

```
##
##      Balanced one-way analysis of variance power calculation
##
##              k = 5
##              n = 15.64184
##              f = 0.4034733
##      sig.level = 0.05
##      power = 0.8
##
## NOTE: n is number in each group
```

To ensure 80% power, you'd need a group size n of at least 16.

- (g) For the sample size determined in the previous part, plot the distribution of F under the null and under the alternative specified by $\omega^2 = 0.14$.

```
library(tidyverse)

n = 16

omega2 = 0.14

cohensf <- sqrt(omega2/(1-omega2))

lambda = n * k * cohensf^2

ggdat <- tibble(x = seq(0, 12, length.out = 10000),
               h0 = df(x = x, df1 = k-1, df2 = k*(n-1)),
               ha = df(x = x, df1 = k-1, df2 = k*(n-1), ncp = lambda))

ggplot(data = ggdat) +
  geom_point(aes(x = x, y = h0, color = "Null")) +
  geom_point(aes(x = x, y = ha, color = "Alternative")) +
  theme_bw() +
  xlab('F') +
  ylab('Density') +
  ggtitle('Null vs. Alternative F Distribution') +
  theme(legend.title=element_blank()) +
  geom_hline(yintercept = 0)
```

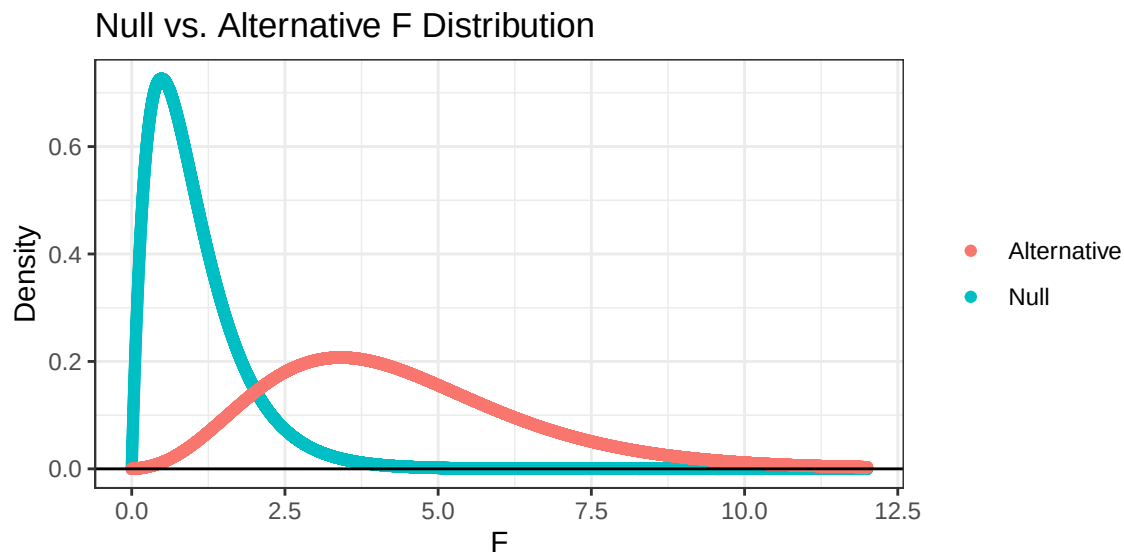


Figure 2: Comparison of F Distributions

- Garofalo et al. (2023) investigated how the immune system communicates with the brain in mice. Specifically, they genetically modified or treated mice to block certain immune cells or signaling pathways.

- The aNK1.1 mice have depleted natural killer (NK) cells, which form an important part of the

immune system.

- The aVLA4 mice have the function of a protein called VLA-4 blocked, which inhibits the effect of NK cells on the brain.
- Rag2^{-/-} mice are genetically modified to lack mature T and B cells, and have NK cells.
- Rag2^{-/-} γ c^{-/-} mice are genetically modified to lack T, B, and NK cells.
- IgG2a and IgG2b are control groups.

In a open-field test, each mouse was placed in an enclosed platform (40 cm $\times$ 40 cm $\times$ 30cm) for 10 minutes. The researchers recorded the distance traveled, the movement duration, and time spent in the center area (20 cm $\times$ 20 cm).

The researchers hypothesized that the mice with inhibited immune system functionality would exhibit increased explorative behavior. This would suggest reduced anxiety-like responses to open spaces. I have provided data about distance traveled in meters, do these data support their hypothesis?

Hint 1: You will have to `pivot_longer()` the data I copied from their posted source data.

Hint 2: Their data do not appear to match the data plotted in their figure for at least the Rag2^{-/-} γ c^{-/-} group. I've emailed the author for updated data (holds breath). Continue as if the data are correct.

```

dat <- read_csv('Garofalo.csv')

dat <- dat |>
  rename(control_1 = IgG2a,
         depleted_nk = aNK1.1,
         control_2 = IgG2b,
         blocked_vla = 'aVLA-4',
         lack_T_B = 'Rag2-/-',
         lack_T_B_nk = 'Rag2-/-gc-/-') |>
  pivot_longer(cols = everything(), values_to = "distance", names_to = "treatment")

norm <- ggplot(data = dat, aes(x = treatment, y = distance)) +
  geom_violin(width = 1) +
  geom_boxplot(width = 0.25) +
  theme_bw() +
  geom_hline(yintercept = 0) +
  xlab('Treatment Group') +
  ylab('Distance Travelled') +
  ggtitle('Distance Travelled by Treatment Group')

logged <- ggplot(data = dat, aes(x = treatment, y = log2(distance))) +
  geom_violin(width = 1) +
  geom_boxplot(width = 0.25) +
  theme_bw() +
  geom_hline(yintercept = 0) +
  xlab('Treatment Group') +
  ylab('Distance Travelled (log2)') +
  ggtitle('Log2 Transformed')

library(patchwork)

norm + logged

```

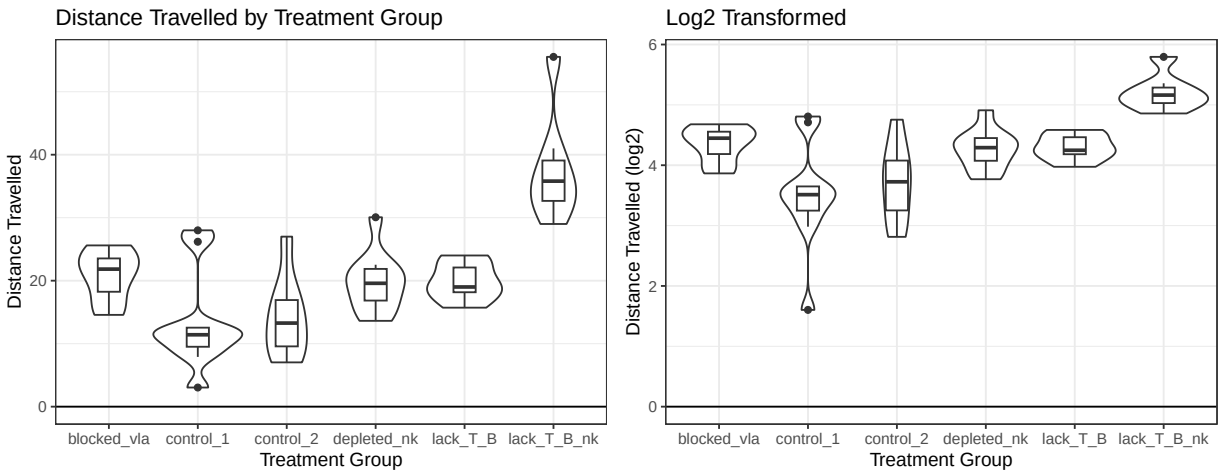


Figure 3: Distance Travelled by Mice Across Treatments

The data appears to be more normally distributed under a $\log_2$ transformation, and it helps with outliers in several of the groups. Also, the variances are clearly not similar across groups so we'll be using a Welch's ANOVA, which does not assume equal variances across groups.

```

library(rstatix)
(aov.test <- welch_anova_test(data = dat, log2(distance) ~ treatment))

## # A tibble: 1 x 7
##   .y.      n statistic    DFn    DFd      p method
## * <chr>   <int>     <dbl> <dbl> <dbl>   <dbl> <chr>
## 1 log2(distance)    60      12.7     5    20.1 0.0000122 Welch ANOVA

```

After performing a Welch's ANOVA on our $\log_2$ transformed data, we obtain a test statistic $F = 12.7$, $p < 0.0001$. Because $p < 0.05 = \alpha$, we have indication that the true population means of distance traveled among the treatment groups are not all equal.

```

dat$log2 <- log2(dat$distance)

(games_howell_test(data = dat, log2 ~ treatment))

## # A tibble: 15 x 8
##   .y. group1 group2 estimate conf.low conf.high p.adj p.adj.signif
## * <chr> <chr> <chr> <dbl> <dbl> <dbl> <dbl> <chr>
## 1 log2 blocked_vla control_1 -0.858 -1.88 0.164 1.2 e-1 ns
## 2 log2 blocked_vla control_2 -0.654 -1.54 0.228 1.89e-1 ns
## 3 log2 blocked_vla depleted_~ -0.0937 -0.575 0.388 9.87e-1 ns
## 4 log2 blocked_vla lack_T_B -0.0570 -0.502 0.388 9.98e-1 ns
## 5 log2 blocked_vla lack_T_B_~ 0.850 0.331 1.37 1 e-3 ***
## 6 log2 control_1 control_2 0.204 -0.971 1.38 9.92e-1 ns
## 7 log2 control_1 depleted_~ 0.764 -0.257 1.79 1.92e-1 ns
## 8 log2 control_1 lack_T_B 0.801 -0.212 1.81 1.51e-1 ns
## 9 log2 control_1 lack_T_B_~ 1.71 0.679 2.74 1 e-3 ***
## 10 log2 control_2 depleted_~ 0.560 -0.321 1.44 3.12e-1 ns
## 11 log2 control_2 lack_T_B 0.597 -0.277 1.47 2.39e-1 ns
## 12 log2 control_2 lack_T_B_~ 1.50 0.614 2.39 2 e-3 **
## 13 log2 depleted_nk lack_T_B 0.0366 -0.400 0.474 1 e+0 ns
## 14 log2 depleted_nk lack_T_B_~ 0.944 0.430 1.46 3.74e-4 ***
## 15 log2 lack_T_B lack_T_B_~ 0.907 0.423 1.39 5.65e-4 ***

```

After performing a Games-Howell pairwise follow-up to Welch's ANOVA, we can see that there are statistically discernible differences of $\log_2$ transformed means between 5 of the pairwise comparisons:

1. Mice with the blocked VLA-4 protein travel greater $\log_2$ transformed distance than mice that lack T, B, and natural killer cells: 95 % CI [0.33, 1.37] ($p = 0.001$).
2. Mice that lack T, B, and natural killer cells travel less $\log_2$ transformed distance than control group 1 mice: 95% CI [-2.74, -0.68] ($p = 0.001$).
3. Mice that lack T, B, and natural killer cells travel less $\log_2$ transformed distance than control group 2 mice: 95% CI [-2.39, -0.61] ($p = 0.002$).
4. Mice that lack T, B, and natural killer cells travel less $\log_2$ transformed distance than mice that have depleted natural killer cells: 95% CI [-1.46, -0.43] ($p < 0.001$).
5. Finally, mice that lack T, B, and natural killer cells travel less $\log_2$ transformed distance than mice that lack T and B cells: 95% CI [-1.39, -0.42] ($p < 0.001$).

References

Garofalo, S., Coccozza, G., Mormino, A., Bernardini, G., Russo, E., Ielpo, D., Andolina, D., Ventura, R., Martinello, K., Renzi, M., et al. (2023). Natural killer cells and innate lymphoid cells 1 tune anxiety-like behavior and memory in mice via interferon- γ and acetylcholine. *Nature Communications*, 14(3103).